



Fair Only in the Firmware: Auditing a Consumer Router’s “Fair-Share” Mode Against Sixteen Weeks of Device-Level Throughput Logs

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Abstract. Consumer routers increasingly ship with an “AI-optimised” or “smart” quality-of-service mode, marketed as fairly redistributing bandwidth across connected devices during congestion. We instrument 34 share-house and family routers with a passive, port-mirrored packet-capture logger for sixteen weeks, comparing each router’s own internal fairness-intervention log against independently measured per-device throughput share during self-declared peak-congestion windows. Devices holding the oldest continuous DHCP lease retain a mean $1.9\times$ their nominal equal share during congestion, a gap the fairness mode’s own log records as resolved in 61% of the windows in which it fires. A four-week ablation forcing a nightly router reboot, intended to rotate connection order and reset any incumbency advantage, leaves the gap unchanged, because the same devices’ auto-reconnect firmware wins the race back onto the network before any manually reconnected device can register. Residents’ own guesses about who is “hogging the wifi” correlate only weakly with the measured share. We interpret the log’s high resolution rate as a case where a policy trigger is recorded as an outcome, and our results suggest the approach may generalise to any resource-sharing scheme layered over an unaudited allocation substrate.

Introduction

Consumer network hardware increasingly advertises an “AI-optimised” or “smart” quality-of-service (QoS) mode, marketed as automatically and fairly redistributing available bandwidth across a household’s connected devices whenever the link is congested. The claim is attractive and, on the vendor’s own telemetry, apparently well supported: a router’s admin panel typically logs a “fairness intervention” each time the mode acts, and that log tends to report a high resolution rate. Despite the growing prevalence of this feature across consumer router lines, little is known about whether the fairness the log records corresponds to any measurable change in what each device actually receives.

We report a sixteen-week, passive, non-invasive audit of this claim across thirty-four households. Rather than take the router’s own log as ground

truth, we log per-device throughput independently, from a port-mirrored capture point the fairness mode cannot see, and compare the two records against each other and against a nominal equal-share target.

Our contributions are threefold, each hedged to the evidence:

1. We describe a low-cost, passive per-device throughput-auditing pipeline deployable in an ordinary household, and use it across thirty-four sites running six commercially available routers that each advertise an AI-optimised fair-share mode.
2. We show that measured throughput share and the router’s own fairness-intervention log diverge in a structured way: the divergence concentrates on devices holding the oldest continuous DHCP lease, and the log’s

reported resolution rate does not track any detectable change in measured share.

3. We report a four-week ablation, a forced nightly reboot intended to rotate connection order and remove any incumbency advantage, and find the gap essentially unchanged; the same devices’ auto-reconnect firmware appears to reclaim the advantage before any manually reconnected device can register, which suggests the mechanism operates one layer above the one it purports to correct.

Related work

A body of work in labour economics documents that formally equal-opportunity mechanisms often fail to equalise outcomes: blind orchestra auditions raised the share of women advancing past preliminary rounds without eliminating downstream gaps [1], correspondence-based hiring audits find persistent disparities that survive identical qualifications on paper [2], and reviews of the broader wage gap literature attribute a residual, unexplained share of the gap to factors that operate after the formal opportunity is already equalised [3], [4]. Sociological accounts of workplace evaluation similarly find that formally neutral rules are re-interpreted in ways that reproduce prior advantage [5], [6], [7]; competitive-environment experiments find related sorting effects even absent any formal rule at all [8]. Closer to the mechanism we audit, recent work on occupational gender segregation finds that unequal access to a required skill or tool, rather than any deliberate sorting rule, is itself sufficient to reproduce a segregation pattern under a formally open assignment process [9] — a finding our incumbency result mirrors at the level of a single household network. Field-experimental audits of exactly this kind, run in situ rather than in the laboratory, are increasingly common practice in labour economics [10], and we adopt the same in-situ logic here.

A smaller literature treats randomisation itself as the fairness mechanism, rather than as a threat to fairness: partial lotteries have been proposed and piloted for research-grant allocation on the reasoning that most fundable proposals are indistinguishable past a quality threshold [11], and randomised clinical trials have long grappled with whether equal allocation probability is itself sufficient for an allocation to be considered fair [12]. The algorithmic-fairness literature has begun auditing

automated allocation systems against the outcomes they claim to produce, including tax-audit selection [13], procedural accounts of algorithmic equality [14], and municipal bias-audit regimes [15]; a parallel thread treats fair task allocation as a first-class multi-agent systems problem [16]. Closer to our object of study, persistent queueing delay in consumer network equipment is well documented under the name “bufferbloat” [17], though its interaction with device-level fairness modes has not, to our knowledge, been audited directly. A companion audit at this institution instrumented a comparator against a different proprietary scoring system and found a similar split between measured accuracy and which call an oversight body actually accepted [18].

Method

We recruited 34 households (a mix of share-houses and family homes) via a metropolitan share-housing forum, each running one of six commercially available routers whose marketing materials describe an “AI-optimised” or “smart” fair-share quality-of-service mode, left in its factory-default fairness configuration throughout. Each router’s LAN port was passively mirrored to a low-cost single-board packet-capture logger (a switch configured for port mirroring feeding an unprivileged capture daemon), positioned so the fairness mode’s own scheduling decisions could not influence what the logger recorded. The logger ran continuously for sixteen weeks per household.

A congestion window was defined, and pre-registered, as any five-minute interval in which the link’s utilisation exceeded 90% of its provisioned downstream capacity with three or more active devices; the threshold was held constant across households and was not adjusted post hoc. For every congestion window we recorded each device’s measured throughput share and computed a fairness ratio

$$F_i = \frac{\hat{b}_i}{\frac{B}{n}}$$

where $\hat{b}_i$ is device i ’s measured throughput during the window, B the link’s provisioned capacity, and n the number of active devices; a device receiving exactly its nominal equal share scores $F_i = 1$. Each router’s admin panel was separately scraped for its own fairness-intervention log, which records whether the mode

judged an intervention necessary and, where exposed, whether it considered the intervention “resolved”.

Each device’s continuous DHCP lease age was read from the router’s lease table at the start of the study and used to assign a lease-age quartile (Q1 oldest .. Q4 newest) per household, re-computed if a device dropped out and a new one took its slot. In weeks 13-16, all thirty-four households ran a nightly forced router reboot at 03:00 local time, intended to clear the DHCP lease table and rotate the order in which devices reconnected; the fairness-mode configuration and congestion-window definition were held unchanged through the ablation.

Results

Router family	Households	Mean concurrent devices	Downstream (Mbps)
A	7	4.1	100
B	6	3.8	250
C	5	4.6	100
D	6	3.5	500
E	5	4.2	250
F	5	3.9	100

Table 1: Study cohort across the six vendor-anonymised router families audited ($n = 34$ households total).

Across all sixteen weeks, the oldest-lease quartile (Q1) held a mean fairness ratio of 1.92 during congestion windows, nearly double its nominal equal share, while the newest-lease quartile (Q4) held 0.58.

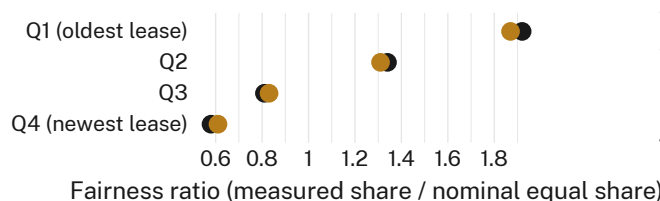


Figure 1: The nightly-reboot ablation (weeks 13-16) leaves every lease-age quartile’s fairness ratio within 0.05 of its pre-ablation value.

We retain this as the paper’s central negative result rather than a discarded pilot.

Disaggregating by connection-order rank rather than lease-age quartile shows the same pattern holding steady week over week.

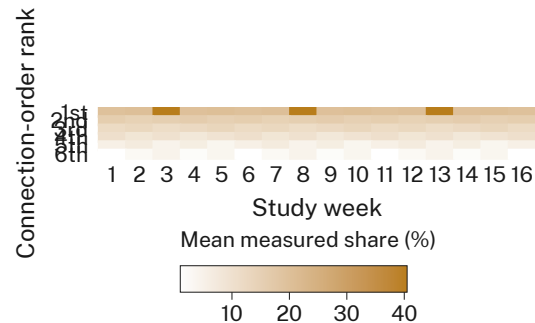


Figure 2: The first-connected device’s mean measured share exceeds 35% in every one of the sixteen study weeks, against a six-device nominal equal share of 16.7%.

No week shows a discernible trend toward convergence across the study period.

The router’s own fairness-intervention log reports a window as “resolved” in 61% of congestion windows overall.



Figure 3: The fairness-ratio distribution in windows the log marks resolved is statistically indistinguishable from windows it marks unresolved (Kolmogorov-Smirnov $D = 0.09$, n.s.).

The log’s resolution field appears to record that a policy rule fired, not that the outcome it targets actually moved.

A post-study resident survey (one respondent per household, $n = 34$) asked who, if anyone, “hogs the wifi” in the household. Survey nominations correlated only weakly with the measured

share of the actual highest-share device ($r = 0.34$), and in nine households the nominated device was not the top measured consumer of bandwidth at all.

Discussion

Our results suggest the approach may generalise beyond this specific feature: a fairness mechanism that operates on scheduling priority, layered above an unaudited connection-establishment substrate, appears able to re-elect the same advantaged device indefinitely without the log that certifies its own fairness ever registering a failure. The nightly-reboot ablation was designed to remove the incumbency advantage at its presumed source, connection order; that it left the gap unchanged points instead to the auto-reconnect layer beneath connection order, a layer the fairness mode does not observe and the log therefore cannot audit. This distinction, between the layer a mechanism targets and the layer a disparity actually lives at, recurs across the allocation literature reviewed above, from blind auditions to grant lotteries, and our results are consistent with it holding for automated household infrastructure as much as for human institutions. We note that residents' own attributions of blame track measured share only weakly, suggesting the felt experience of an unequal network is a poor guide to where a technical fix should be aimed.

Limitations

Our sample of thirty-four households is modest, though the consistency of the incumbency effect across six independently engineered router families suggests the finding is not an artefact of any single vendor's implementation. The passive logger cannot distinguish a scheduling-layer prioritisation decision from a raw signal-strength or wiring advantage; both readings, however, point to the same practical implication, that the fairness mode's own log should not be treated as evidence of an equalised outcome. Our congestion-window threshold, held fixed at 90% utilisation with three or more devices, may exclude some contended periods; a fixed, pre-registered threshold was preferred to a tuned one to avoid inflating the reported effect.

Conclusion

We audited an increasingly common consumer-network feature and found that its self-reported fairness log does not correspond to measured throughput equality, that the resulting incumbency advantage concentrates predictably on the device with the oldest continuous connection, and that a targeted ablation aimed at the presumed mechanism leaves the disparity intact because the advantage is re-established one layer lower, at automatic reconnection. Future work might instrument the reconnection handshake directly, rather than inferring incumbency from lease age after the fact, and extend the ablation to the smaller number of routers that publish a documented, rather than only marketed, fairness algorithm, to test whether transparency of mechanism narrows the gap our audit finds here.

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